

Abdelhakim Amer

MSc. in Mechanical Engineering from TU Delft
PhD in Robotics and A.I from Aarhus University
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SOCIAL LINKS

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ACADEMICS

I'm a postdoctoral researcher at Aarhus University, where I also completed my PhD in learning-based optimal control. I hold bachelor's degree in Mechanical Engineering (with High Honors) and a master's from Delft University of Technology. Also I was a visiting scholar at the German Center for Artificial Intelligence (DFKI).



	2022 - 2025	EIVA A/S, Denmark	PhD , Robotics and A.I
	2025 (3 month)	German Research Center for Artificial Intelligence, Germany	Visiting Scholar , Underwater Robotics Research Group
	2023 (1 month)	University of Paderborn, Germany	Visiting Scholar , Automatic Control Research Group
	2018 - 2021	Delft University of Technology, The Netherlands	Master , Mechanical Engineering
	2013 - 2018	The American University in Cairo, Egypt	Bachelor , Mechanical Engineering (High Honors)

WORK EXPERIENCE

After graduation, I spent a year at the AIR Lab at Aarhus University, working on an autonomous wind turbine inspection project in collaboration with Upteko. I then started my industrial PhD at EIVA, where I did my research in learning-based optimal control, having developed novel methods for planning and control of autonomous underwater vehicles.

	2022 - 2025	EIVA A/S	PhD Researcher in Robotics
	2021 - 2025	Aarhus University	Graduate Teaching Assistant in Control and Robotics
	2021 - 2022	Upteko Aps	R&D Engineer
	2021 - 2022	AIR Lab, Aarhus University	Research Assistant in Autonomous Systems
	2020 (3 months)	Royal IHC	Dynamics and Control Graduate Intern

PROGRAMMING EXPERIENCE



Python
★★★★★



C++
★★★★★



ROS
★★★★★



Unreal Engine
★★★★★



MATLAB
★★★★★



LaTeX
★★★★★



Shell Scripting
★★★★★

TEACHING AND SUPERVISION

As a Teaching Assistant at Aarhus University, I contributed to the following courses:

- Autonomous Mobile Robots
- Control Theory

Additionally, I have supervised 10 Bachelor's and Master's theses, guiding students through projects in areas such as robot modelling and advanced control.

FUNDING AWARDS

I have received the following funding awards during my studies:

 **Innovation Fund Denmark (1.1M DKK)** (2022-2025) - (BIO-CLEAN) AI-powered marine robotics .

 **Academic Excellence Scholarship** (2013-2018) – Awarded for outstanding academic performance.

PUBLICATIONS

The following list of publications showcases the outcomes of my research. I have published in leading robotics and control journals and conferences, collaborating with researchers from top institutions and companies worldwide, including Denmark, Germany, Finland, China, and the USA.

1. **A. Amer** , A. Alstrup, F. Toft, Y. Brodskiy, A. Sarabakha, and E. Kayacan, *SeaVis: Modeling and Control of a Remotely Operated Towed Vehicle for Seabed Visualization and Mapping*, "under review" at the *European Control Conference (ECC)*, 2026.
2. **A. Amer** and A. Sarabakha, *Dynamic System Emulation: Fixed Wing Dynamics on a Multicopter*, "under review" at the *IEEE Conference on Control Technology and Applications (CCTA)*, 2026.
3. **A. Amer**, M. Mehndiratta, Y. Brodskiy, and E. Kayacan, *REACT: Real-Time Entanglement-Aware Coverage Path Planning for Tethered Underwater Vehicles*, "accepted" at the *IEEE International Conference on Robotics & Automation (ICRA)*, 2026.
4. **A. Amer**, M. Mehndiratta, Y. Brodskiy, and E. Kayacan, *Empowering Autonomous Underwater Vehicles Using Learning-Based Model Predictive Control With Dynamic Forgetting Gaussian Processes*, *IEEE Transactions on Control Systems Technology*, early access, 2025, pp. 1–8. doi: 10.1109/TCST.2025.3539218.
5. W. Liang, **A. Amer**, M. Mehndiratta, Z. Chen, B. Yao, and E. Kayacan, *Adaptive Robust Control Integrated With Gaussian Processes for Quadrotors: Enhanced Accuracy, Fault Tolerance and Anti-Disturbance*, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 55, no. 5, pp. 3235–3248, May 2025. doi: 10.1109/TSMC.2025.3539707.
6. **A. Amer**, D. Falsegar, Y. Brodskiy, and A. Sarabakha, *Modelling of Underwater Vehicles Using Physics-Informed Neural Networks With Control*, accepted for publication in the *International Joint Conference on Neural Networks (IJCNN)*, 2025. Available at: <https://arxiv.org/abs/2504.20019>
7. **A. Amer**, M. Mehndiratta, J. L. F. Sejersen, H. X. Pham, and E. Kayacan, *Visual Tracking Nonlinear Model Predictive Control Method for Autonomous Wind Turbine Inspection*, in *Proceedings of the 2023 21st International Conference on Advanced Robotics (ICAR)*, pp. 431–438, 2023. doi: 10.1109/ICAR.2023.10406329.
8. **A. Amer**, O. Álvarez-Tuñón, H. İ. Uğurlu, J. L. F. Sejersen, Y. Brodskiy, and E. Kayacan, *UNav-Sim: A Visually Realistic Underwater Robotics Simulator and Synthetic Data-Generation Framework*, in *Proceedings of the 2023 21st International Conference on Advanced Robotics (ICAR)*, pp. 570–576, 2023. doi: 10.1109/ICAR.2023.10406819.

REFERENCES / RECOMMENDATIONS

The following individuals can provide recommendations on my academic, and professional work:

Prof. Erdal Kayacan, Full Professor, University of Paderborn, Germany – erdal.kayacan@uni-paderborn.de

Prof. Andriy Sarabakha, Associate Professor, Aarhus University, Denmark – andriy@ece.au.dk

Dr. Bilal Wehbe, Underwater Robotics Group Lead, DFKI, Germany – bilal.wehbe@dfki.de

Dr. Yuri Brodskiy, R&D Lead, EIVA A/S, Denmark – ybr@eiva.com